Implementing and Evaluating a Simple AI Model: F1 DNF Prediction

This study implements a Random Forest binary classifier to predict whether a Formula 1 driver will fail to finish a race (DNF — Did Not Finish), using historical race data from 1990 to the present. The problem is framed as a binary classification task: given pre-race information such as grid position, constructor identity, and historical reliability rates, the model predicts whether a driver will retire before the race ends. Accurate DNF prediction has practical applications in race strategy, resource planning, and safety decision-making.

The model was trained on 14,086 records drawn from the Rohanrao Formula 1 dataset, with a DNF rate of 32.10%, confirming meaningful class imbalance. Overall accuracy reached 69.98%; however, this figure is insufficient as a standalone metric. Precision of 52.94% indicates that just over half of predicted DNFs were genuine, while recall of 57.74% shows that the model identified fewer than three in five actual retirements. The confusion matrix shows 464 false positives and 382 false negatives—a trade-off that would carry real consequences in a production safety context. Feature importance analysis identified the constructor DNF rate as the strongest predictor, followed by year and driver DNF rate, confirming that team reliability and era are more predictive than grid position alone.

Several ethical considerations warrant attention. The dominance of the constructor DNF rate as a feature means the model encodes historical team performance, which could unfairly reflect on constructors that have since improved their reliability. Additionally, data from before 1994 — prior to the safety reforms following Ayrton Senna's death — reflects a period when retirement rates were shaped by fundamentally different safety standards, which risks skewing predictions for modern conditions. Any use of this model in a safety-critical context would require regular retraining to reflect current regulations and technology.

Metric	Value
Accuracy	~70%
Precision (DNF)	~53%
Recall (DNF)	~58%